

# compute\_spatial\_mev\_basis.R

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| compute_spatial_mev_basis                   |
| <i>Compute a Reusable Spatial MEM Basis</i> |

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## Description

Computes exact or low-rank Moran eigenvectors and retains explicit state required to project the same basis to held-out locations.

## Usage

```
compute_spatial_mev_basis(  
  data_coords_projected = NULL,  
  n_mev = 20L,  
  strategy = "exact",  
  exact_max_locations = 1999L,  
  fast_eigenvectors = 200L,  
  fast_seed = 900723L,  
  exact_function = sjSDM::generateSpatialEV,  
  fast_function = NULL  
)
```

## Arguments

|                                                  |                                                                             |
|--------------------------------------------------|-----------------------------------------------------------------------------|
| data_coords_projected                            | Data frame with 'coord_x_km' and 'coord_y_km' columns and unique row names. |
| n_mev                                            | Positive integer giving the number of public eigenvectors requested.        |
| strategy, exact_max_locations, fast_eigenvectors | Shared strategy settings validated by [resolve_spatial_mev_strategy()].     |
| fast_seed                                        | Positive integer used locally for deterministic NystrC6m knot selection.    |
| exact_function                                   | Exact Moran-eigenvector function. Defaults to [sjSDM::generateSpatialEV()]. |
| fast_function                                    | Fast Moran-eigenvector function. Defaults to [spmoran::meigen_f()].         |

**Details**

Exact results retain the current sjSDM values unchanged. Fast results use an exponential-kernel basis and orient each selected eigenvector deterministically. The orientation is retained for matching projection.

**Value**

Named list with 'data\_mev', 'projection\_basis', and one-row 'data\_provenance'.

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